JUAN DIEGO DÍAZ AVENDAÑO

Backend Developer | Java, Spring Boot, Node.js, REST APIs, PostgreSQL

Montería, Colombia (Remote) | +57 305 298 0337 | juandiegodiazave@gmail.com | linkedin.com/in/juandiegodiaz-dev | github.com/juandda0

PROFESSIONAL SUMMARY

Backend developer with 3 years of experience building REST APIs, microservices and SaaS products with Java and Spring Boot. Delivered production systems for a university, a demolition company and a dental practice using PostgreSQL, Spring Security (JWT), Spring Data JPA, Hibernate, Flyway, Docker and Node.js, plus serverless backends on Firebase Cloud Functions. Comfortable owning a feature from requirements to deployment. Winner of TechU Montería 2025 (Montería City Hall) and 1st place in the Universidad de Córdoba Systems Engineering Hackathon. Available immediately for full-time remote work.

TECHNICAL SKILLS

Languages: Java, TypeScript, JavaScript, SQL

Backend: Spring Boot, Spring Security, Spring Data JPA, Hibernate, REST APIs, Microservices, JWT, Node.js, Webhooks, Zod

Databases: PostgreSQL, Firestore, Flyway, data modeling, database migrations, optimistic locking

Cloud & DevOps: Docker, Docker Compose, Nginx, Firebase (Auth, Firestore, Cloud Functions, Hosting, App Check), Railway, Vercel, Git, GitHub, CI/CD

Testing: JUnit, Jest, Testing Library, Firebase Emulator tests, Maestro E2E

Frontend & Mobile: React, ReactJS, React Native, Expo, TypeScript

Integrations: Wompi payments, RevenueCat subscriptions, AI/LLM APIs, S3-compatible storage

WORK EXPERIENCE

Full-Stack Developer (Backend Focus) | Universidad de Córdoba

Montería, Colombia (Remote) | Aug 2025 – Jun 2026

- Built SIGCORP end to end, an internal request-management platform used by 17 staff members to process about 40 requests per day: REST APIs, assignment workflows, status tracking, role-based access and dashboards.
- Designed security and persistence with Java, Spring Boot, Spring Security, JWT, Spring Data JPA, Hibernate, Flyway and PostgreSQL; used optimistic locking to keep data consistent in concurrent workflows.
- Containerized and deployed the platform with Docker, Docker Compose and Nginx on institutional infrastructure, from requirements to maintenance.
- Built backend modules for an institutional e-commerce platform: Wompi payment gateway with asynchronous webhook confirmation and a Node.js notification microservice using REST service-to-service communication.
- Contributed to the migration of about 200 pages of the university website; worked with Git, GitHub, pull requests, code review and Scrum.

FREELANCE EXPERIENCE

Backend Developer & Team Lead | Sonrisa – Dental Practice Management SaaS

Remote | Aug 2026 – Oct 2026

- Designed and built the Java 21 / Spring Boot and PostgreSQL backend for patients, appointments, 19-step clinical records with electronic signature, odontogram, treatment plans, payments and finance.
- Implemented Spring Security with JWT and refresh tokens, JPA/Hibernate and Flyway migrations; immutable signed records and an access audit log to comply with Colombian health-data regulations (Ley 1581, RIPS, CIE-10, CUPS).
- Led a team of three: gathered requirements directly with the client, defined the architecture, reviewed pull requests and integrated the React + TypeScript frontend; deployed with Docker on Railway and Vercel.

Full-Stack Developer | SGO – ESCAMVIOS S.A.S.

Remote | May 2026 – Jun 2026

- Replaced an Excel-based transportation workflow with a React, TypeScript and Firebase web application for trips, projects, employees, attendance and client access.
- Migrated and structured 550+ historical trip records; built KPI dashboards, charts, filters, search and pagination.
- Implemented Firestore Security Rules with role-based and project-level data isolation, write validation and emulator-based security tests; first designed the backend in Spring Boot and PostgreSQL, then migrated it to Firebase to fit the client's scale and budget.

Full-Stack Mobile Developer | CETI – AI Financial Education Platform

Remote | Apr 2025 – Sep 2025

- Built a cross-platform mobile and web app with React Native, Expo, TypeScript and Firebase: gamified lessons, quizzes, streaks, savings goals, expense tracking, budgets and charts.
- Built a server-side AI workflow on Firebase Cloud Functions that turns natural-language and voice requests into structured actions, with validation, explicit user confirmation, usage quotas and local fallbacks.
- Implemented Firebase Auth, Custom Claims and Firestore Security Rules isolating data by school, course and user; added automated tests with Jest, Testing Library, Firestore emulator tests and Maestro E2E.

Full-Stack Mobile Developer | Biomech – B2B Fitness SaaS
Remote | May 2023 – Jun 2025

- Built a two-sided mobile app (trainers and athletes) with React Native, Expo and TypeScript: clients, routines, progress tracking, weekly streaks, scheduling, bookings and payments. Published on the App Store.
- Designed a serverless Firebase backend with 30+ protected Cloud Functions, Zod validation, Firebase App Check and Firestore/Storage Security Rules, with multi-organization support for locations, trainers and clients.
- Integrated Wompi payments and RevenueCat subscriptions for App Store and Android billing.

AWARDS & RECOGNITION

- **Winner – TechU Montería 2025, Montería City Hall (May 2025).** CTO of Unimarket, a university marketplace with catalogs, auctions, raffles and its own digital currency (UNICOIN); one of 5 winning projects out of 72 proposals.
- **1st place – Systems Engineering Hackathon, Universidad de Córdoba (May 2025).** AulaLink, an attendance-tracking system for schools, built in a team of three: Express.js and TypeScript backend, Tailwind CSS web SPA, Flutter mobile app, and SQL and NoSQL databases.
- **Winner – Innovation Fair, Universidad de Córdoba (May 2025).** Unimarket, as CTO of a four-student team from the Innovarium research group.

INTERNATIONAL CONFERENCE PRESENTATIONS

- **“LUMIO: Automated Financial Management and Education” (lead author).** 8th CIIDIES–COPEIN–RIISS International Congress and 10th Emprende Innova UG Congress, Celaya, Mexico (virtual), Oct 2026.
- **“CETI: Financial Education Platform” (co-author).** Same congress, Celaya, Mexico (in person), Oct 2026.
- **“UNIMARKET” (co-author).** 5th International Congress on Technology, Learning and Education (CITAE 2025), REDIIE, Santiago, Chile (hybrid), Jul 2025.

EDUCATION

B.S. in Systems Engineering (in progress, 7th semester) | Universidad de Córdoba
Montería, Colombia | 2023 – Expected 2027

Member of the Innovarium research group (Faculty of Engineering).

LANGUAGES

Spanish – Native | English – C1 (Professional)